INCH-POUND

MIL-W-15805F (AS)
AMENDMENT 4
30 April 2001
SUPERSEDING
AMENDMENT 3
18 October 1977

MILITARY SPECIFICATION

WIND DIRECTION AND WIND SPEED INDICATING EQUIPMENT, TYPE B

This amendment forms a part of MIL-W-15805F (AS), dated 10 December 1962, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: STANDARDS, MILITARY: Delete reference to "MIL-STD-242" and add to the following:

"MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-740	-	Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment."

PAGE 2

3.1: Delete and substitute:

* "3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection in accordance with 4.2."

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PAGES 2 and 3

3.3.1: Column 1, lines 13 and 14: delete; In column 2, line 4: delete “Iron or steel” and substitute “Ferrous alloy;” line 31: add “and inductors,” line 33: delete “see” and substitute “except;” line 46: delete; line 47: delete “and circuits,” line 50: add “(see 3.4.1, 3.4.2, 3.4.3.1.1, 3.4.3.2.1, 3.4.4.2.1 and 3.4.6).” In column 3, line 5: delete “Welding” and substitute “Structural welding;” line 9: add “(not required for transmitter and headwind computer);” line 14: delete “Personnel protection” and substitute “Safety (personnel hazard);” line 15: add “(see 3.3.1.3);” line 22: delete; line 30: add “(see 3.3.1.4);” line 48: delete “and general examination.”

PAGE 3

3.3.1.1: Delete.

Add new paragraphs:

“ 3.3.1.3 Electromagnetic interference emission and susceptibility. The equipment shall conform to the electromagnetic interference emission and susceptibility requirements for class ID of MIL-STD-461.

3.3.1.4 Structurebone noise. The equipment shall conform to the structurebone noise acceptance level for type 3 of MIL-STD-740, except the correction factor “for solidly mounted equipment subtract 20DB from the curves” is not acceptable.”

3.3.2.1: In item (b), delete “45” and substitute “46,” in item (c), delete “60” and substitute “061;” and in item (d), delete “80” and substitute “81.”

3.4.1, line 15: Delete “splashproof” and substitute “115-volt.”

PAGE 4

3.4.2, line 24: Delete “120-volt” and substitute “115-volt”.

PAGE 5

3.4.3, 3.4.3.1, and 3.4.3.2: Delete and substitute:

“3.4.3 Wind direction and speed indicator. The type B wind direction and wind speed indicator equipment shall be furnished with type B or type F wind direction and speed indicators as specified (see 6.1).

3.4.3.1 Type B.

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3.4.3.1.1 Wind indicator. The type B wind indicator shall be manufactured in accordance with drawing 3331241. The Indicator shall contain, in a watertight enclosure, the mechanism incorporating size 1F synchros, as specified in MIL-S-2335, to receive signals from the wind direction and speed transmitter, and to indicate the relative wind direction in degrees and the apparent wind speed in knots. The indicator shall contain two independent units, one for direction and the other for speed. Electrical connections shall be made to each unit by plug-jack type terminals, Winchester types A7P and A7S, or equal. The units shall be bolted in the case. A hole in the case and plug with a 1 inch National Pipe Thread Class 2 in accordance with Handbook H-28 shall be provided in one end for a stuffing tube fitting.

3.4.3.1.2 Size. Type B indicators shall be as small and lightweight as practicable, but in no case shall dimensions and weight exceed the following:

Height – 13.5625 inches
Width – 7.50 inches
Depth – 5.875 inches
Weight - 23 pounds, 7 ounces
(Figure 4 is provided for guidance only)

3.4.3.1.3 Dials. Type B indicator shall have two dials 4 inches in diameter; one dial shall indicate wind direction and the other shall indicate wind speed. The dials shall be marked in accordance with drawing 3331269. Red dial illumination shall be provided in the indicator.

3.4.3.2 Type F.

3.4.3.2.1 Wind indicator. The type F wind indicator shall be manufactured in accordance with Bendix Corporation, Friez Instrument Division drawing 1148595. The indicators shall contain, in a watertight enclosure, the mechanism incorporating type 18TR6 synchro in accordance with MIL-S-20708, to receive signals from the wind direction and speed transmitter and to indicate the relative wind direction in degrees and the apparent wind speed in knots. The indicator shall contain two synchros on a single chassis. Electrical connections shall be made to the case by a plug and jack connector; by an Amphenol 26-4100-16P and Amphenol 26-4200-16s; or equal. A hole at each end of the case, size 1 inch National Pipe Thread Class 2 in accordance with Handbook H-28, shall be provided for cable entrance. A power transformer for the lighting circuit shall be located in the housing. A variable resistor for controlling the light intensity shall be provided and adjusted by a knob from the outside. The cover of the housing shall provide a single glass for observing the two dials.

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3.4.3.2.2 Size. Type F indicators shall be as small and lightweight as practicable, but in no case shall dimensions and weight exceed the following:

Height – 11.125 reference inches
Width – 7.375 reference inches
Depth – 4.8125 reference inches
Weight - 15 pounds
(Figure 9 is provided for guidance only)

3.4.3.2.3 Dials. Type F indicators shall have two dials 4.230 ± 0.010 inches in diameter; one dial shall indicate wind direction and the other shall indicate wind speed. The dials shall be marked in accordance with drawing 3331269. Red dial illumination shall be provided in the indicator."

3.4.3.6, after "Illumination:" Add "(Type B)."

3.4.4.2, line 2: Delete "bound" and substitute "mounted," and add "dripproof" after "aluminum."

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3.4.4.15, line 5: Delete "120 volts" and substitute "115 volts."

Add:

"3.4.4.20 Shock and vibration. The wind direction and wind speed recorder shall not be subjected to the shock and vibration requirements specified in MIL-I-983."

PAGE 7

3.4.5.3, line 4: Delete "12" and substitute "27-43."

3.4.7: Delete and substitute:

"3.4.7 Repair parts. Repair parts and applicable repair parts lists required for maintenance of the parent equipment shall be in accordance with table I."

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Table I: Delete and substitute:

"Table I. Onboard repair parts required per ship.

Item	Part	All Others	CVA's	ASROC equipped
		MOD 0	MOD 0	MOD 0
	Detector, transmitter indicator			
1	Ball bearings (one of each type used in the detector) (only if nonstandard)	1	1	1
2	Pliers "tru-arc" (types 62562-A and 60464)	1	1	1
3	Dampener assembly (transmitter)	0	1	1
4	Integrator assembly (transmitter)	0	1	1
5	Box special screws (screws not available in normal stock)	1	1	1
6	Roller (disc), shaft assembly	1	1	1
	Crosswind and headwind computer and indicator			
7	Potentiometer (one of each)	0	1	0
8	Motor assembly	0	1	0
9	Microammeter movement	0	1	0

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* 4.2: Delete "Qualification" and substitute "First article" (3 places).

* 4.2 1/: Delete.

PAGE 9

4.4: In group B, add the following test after "supply line voltage and frequency:"

"Test	Requirement paragraph	Test paragraph
Dielectric strength and insulation resistance	3.3.1	MIL-I-983"

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4.4: In group C, after “airborne noise,” add “(grade C);” add the following test after “airborne noise:”

“Test	Requirement paragraph	Test paragraph
Structureborne noise (type 3)	3.3.1 and 3.3.1.4	4.5.8”

4.4: Group C, opposite “shielding and radio frequency noise reduction,” under test paragraph, delete “MIL-I-983” and substitute “4.5.7.”

Add the following test after “mounting:”

“Test	Requirement paragraph	Test paragraph
Enclosure	3.3.1	MIL-I-983”

PAGE 10

4.4: Group C, “shock” test: before “recorder” insert “detector or.”

PAGE 11

4.5.1.2(b) (4)c, lines, 20, 21, 22, and 23: Delete and substitute:

“r.p.m., 75.6 knots.
57 cycles/1800
r.p.m.
(60 of 75.6) 80 knots.”
57

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Table III, second column: Delete and substitute:

Correct intensity Knots
16.8
33.6
50.4
67.2
75.6

Add the following paragraphs:

"4.5.7 Electromagnetic interference emission and susceptibility. Electromagnetic interference emission and susceptibility tests shall be performed in accordance with MIL-STD-462 to determine conformance with 3.3.1.3.

4.5.8 Structureborne noise. Structureborne noise tests shall be performed in accordance with type 3 of MIL-STD-740 to determine conformance with 3.3.1.4."

5.1 and 5.2.1.1: Add "(see 5.3)."

Add the following paragraphs:

"5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labeled as follows:

CAUTION

Contents cushioned, etc. with polystyrene (loose-fill) material
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material.

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5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level packing and level A and B packing applications such as cushioning, filler and dunnage."

PAGES 14

6.1: Add "(h) whether type B or type F indicator is required (see 3.4.3).
(i) when polystyrene (loose-fill) material is approved (see 5.3)."
* (j) when first article is required (see 3.1)."

* 6.2: Delete.

* 6.3: Delete.

PAGES 28 and 29

Insert attached figure 9.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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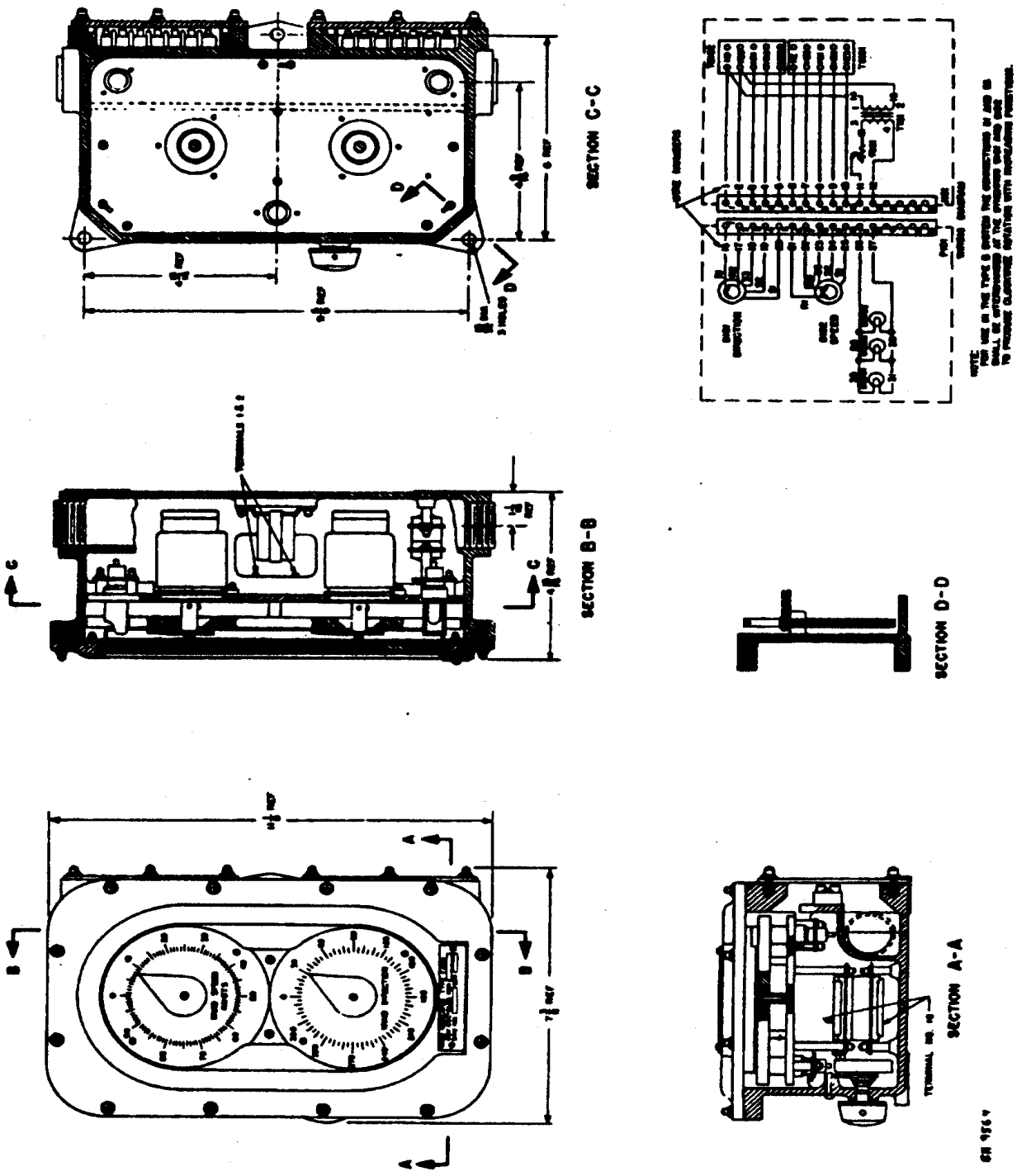


FIGURE 9. Wind direction and speed indicator, Type F.

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